ETSI EN 300 019-1-2 V3.1.1 (2026-09)



EUROPEAN STANDARD

Environmental Engineering (EE);
Environmental conditions and environmental tests for telecommunications equipment;
Part 1: Classification of environmental conditions;
Subpart 2: Transportation

Reference
REN/EE-017018

Keywords
environment, equipment practice, testing

ETSI

650 Route des Lucioles
F-06921 Sophia Antipolis Cedex - FRANCE

Tel.: +33 4 92 94 42 00 Fax: +33 4 93 65 47 16

Siret N° 348 623 562 00017 - APE 7112B
Association à but non lucratif enregistrée à la
Sous-Préfecture de Grasse (06) N° w061004871

Important notice

The present document can be downloaded from the
ETSI Search & Browse Standards application.

The present document may be made available in electronic versions and/or in print. The content of any electronic and/or print versions of the present document shall not be modified without the prior written authorization of ETSI. In case of any existing or perceived difference in contents between such versions and/or in print, the prevailing version of an ETSI deliverable is the one made publicly available in PDF format on ETSI deliver repository.

Users should be aware that the present document may be revised or have its status changed, this information is available in the Milestones listing.

If you find errors in the present document, please send your comments to the relevant service listed under Committee Support Staff.

If you find a security vulnerability in the present document, please report it through our Coordinated Vulnerability Disclosure (CVD) program.

Notice of disclaimer & limitation of liability

The information provided in the present deliverable is directed solely to professionals who have the appropriate degree of experience to understand and interpret its content in accordance with generally accepted engineering or other professional standard and applicable regulations.
No recommendation as to products and services or vendors is made or should be implied.
In no event shall ETSI be held liable for loss of profits or any other incidental or consequential damages.

Any software contained in this deliverable is provided "AS IS" with no warranties, express or implied, including but not limited to, the warranties of merchantability, fitness for a particular purpose and non-infringement of intellectual property rights and ETSI shall not be held liable in any event for any damages whatsoever (including, without limitation, damages for loss of profits, business interruption, loss of information, or any other pecuniary loss) arising out of or related to the use of or inability to use the software.

Copyright Notification

No part of this document may be reproduced in any form, by any means and in any media, without the prior written authorization of ETSI and except as expressly permitted below.
By way of exception and when the document is a normative deliverable (European Standard (EN), Technical Specification (TS), Group Specification (GS) or ETSI Standard (ES)), ETSI authorizes to reproduce and incorporate into products, services and technical documentation only those extracts (e.g. templates) that are strictly necessary for the technical implementation of the normative deliverable, to ensure compliance with the latter.
Nothing in this notice shall be construed as limiting any mandatory exceptions to copyright provided by applicable law.

© ETSI 2026.
All rights reserved.

Contents

Intellectual Property Rights 4
Foreword 4
Modal verbs terminology 5
1 Scope 6
2 References 6
2.1 Normative references 6
2.2 Informative references 6
3 Definition of terms, symbols and abbreviations 7
3.1 Terms 7
3.2 Symbols 7
3.3 Abbreviations 7
4 Environmental classes 7
4.0 Introduction 7
4.1 Class 2.10: Weather protected transportation with temperature control and no humidity control 7
4.2 Class 2.11: Weather protected transportation with no temperature control and humidity control 7
4.3 Class 2.12: Non-weather protected transportation 8
5 Environmental conditions 8
5.1 Climatic conditions 8
5.2 Biological conditions 9
5.3 Chemically active substances 9
5.4 Mechanically active substances 9
5.5 Mechanical conditions 10
Annex A (informative): Bibliography 11
Annex B (informative): Change history 12
History 13

Intellectual Property Rights

Essential patents

IPRs essential or potentially essential to normative deliverables (European Standard (EN), Technical Specification (TS), Group Specification (GS) or ETSI Standard (ES)) may have been declared to ETSI. The declarations pertaining to these essential IPRs, if any, are publicly available for **ETSI members and non-members**, and can be found in ETSI SR 000 314: *"Intellectual Property Rights (IPRs); Essential, or potentially Essential, IPRs notified to ETSI in respect of ETSI standards"*, which is available from the ETSI Secretariat. Latest updates are available on the ETSI IPR online database.

Pursuant to the ETSI Directives including the ETSI IPR Policy, no investigation regarding the essentiality of IPRs, including IPR searches, has been carried out by ETSI. No guarantee can be given as to the existence of other IPRs not referenced in ETSI SR 000 314 (or the updates on the ETSI Web server) which are, or may be, or may become, essential to the present document.

Trademarks

The present document may include trademarks and/or tradenames which are asserted and/or registered by their owners. ETSI claims no ownership of these except for any which are indicated as being the property of ETSI, and conveys no right to use or reproduce any trademark and/or tradename. Mention of those trademarks in the present document does not constitute an endorsement by ETSI of products, services or organizations associated with those trademarks.

DECT™, **PLUGTESTS™**, **UMTS™** and the ETSI logo are trademarks of ETSI registered for the benefit of its Members. **3GPP™**, **LTE™** and **5G™** logo are trademarks of ETSI registered for the benefit of its Members and of the 3GPP Organizational Partners. **oneM2M™** logo is a trademark of ETSI registered for the benefit of its Members and of the oneM2M Partners. **GSM**® and the GSM logo are trademarks registered and owned by the GSM Association.

Foreword

This European Standard (EN) has been produced by ETSI Technical Committee Environmental Engineering (EE).

The present document is part 1, sub-part 2 of a multi-part deliverable covering the classification of environmental conditions and environmental tests for telecommunications equipment, as identified below:

Part 1: "Classification of environmental conditions": (see note 1)

Sub-part 0: "Introduction";

Sub-part 1: "Storage";

Sub-part 2: "Transportation";

Sub-part 3: "Stationary use at weatherprotected locations";

Sub-part 4: "Stationary use at non-weatherprotected locations";

Sub-part 5: "Ground vehicle installations";

Sub-part 6: "Ship environments";

Sub-part 7: "Portable and non-stationary use";

Sub-part 8: "Stationary use at underground locations";

Part 2: "Specification of environmental tests" (see note 2).

NOTE 1: Specifies different standardized environmental classes covering climatic and biological conditions, chemically and mechanically active substances and mechanical conditions during storage, transportation and in use. Sub-part 1-0 forms a general overview of part 1.

NOTE 2: Specifies the recommended test severities and test methods for the different environmental classes.

National transposition dates	
Date of adoption of this EN:	21 September 2026
Date of latest announcement of this EN (doa):	31 December 2026
Date of latest publication of new National Standard or endorsement of this EN (dop/e):	30 June 2027
Date of withdrawal of any conflicting National Standard (dow):	30 June 2027

Modal verbs terminology

In the present document "**shall**", "**shall not**", "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the ETSI Drafting Rules (Verbal forms for the expression of provisions).

"**must**" and "**must not**" are **NOT** allowed in ETSI deliverables except when used in direct citation.

1 Scope

The present document defines classes of environmental conditions and their severities to which telecommunication equipment may be exposed. The severities specified are those which will have a low probability of being exceeded; generally less than 1 % of the time of a transport.

The present document applies to equipment being transported from one place to another after being made ready for dispatch from the manufacturer's works. The most commonly used methods of transportation have been taken into account, i.e. ground, water and air transport. Loading and unloading as well as temporary storage, have been included. Where the equipment is packaged the environmental conditions apply to the packaged equipment.

NOTE: Normal transportation time is considered to be 30 days or less. Where the total transportation time exceeds 30 days then additional storage or packaging precautions should be considered.

The tests applicable to verify the compliance with the transportation and handling environmental classes of the present document, are defined in ETSI EN 300 019-2-2 [i.5].

2 References

2.1 Normative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

Referenced documents which are not found to be publicly available in the expected location might be found in the ETSI docbox.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, ETSI cannot guarantee their long-term validity.

The following referenced documents are necessary for the application of the present document.

Not applicable.

2.2 Informative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, ETSI cannot guarantee their long-term validity.

The following referenced documents may be useful in implementing an ETSI deliverable or add to the reader's understanding, but are not required for conformance to the present document.

[i.1] Void.

[i.2] IEC 60721-3-2:2018: "Classification of environmental conditions - Part 3-2: Classification of groups of environmental parameters and their severities - Section 2: Transportation and handling".

[i.3] Void.

[i.4] Void.

[i.5] ETSI EN 300 019-2-2: "Environmental Engineering (EE); Environmental conditions and environmental tests for telecommunications equipment; Part 2-2: Specification of environmental tests; Transportation".

3 Definition of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the following terms apply:

absolute humidity: mass of water vapour in grammes which is associated with one cubic metre of dry air in an air/water vapour mixture

non-weatherprotected location: equipment, packaged or unpackaged, is not protected in any way from the environment

relative humidity: ratio of the partial pressure of the water vapour in moist air at a given temperature, to the partial pressure of the water vapour in saturated air at the same temperature

weatherprotected location: location at which the equipment is protected from direct weather influences

NOTE: Totally weatherprotected location: the equipment, packaged or unpackaged, is contained within an enclosure which affords some protection from the environment, ranging from a temperature controlled container to a waterproof cover placed over the equipment. Ventilation ranges from controlled air flow to the raising of part of a waterproof cover to allow for natural air flow.

3.2 Symbols

Void.

3.3 Abbreviations

Void.

4 Environmental classes

4.0 Introduction

Three different environmental classes have been defined. Classes 2.10 and 2.11 are classes for weather-protected transportation with less severe mechanical conditions. Class 2.12 is the class for non-weather protected transportation.

NOTE: The classes defined in previous versions of the present document have been replaced with new classes as a result of recent efforts at collecting information regarding climatic conditions performed in IEC.

4.1 Class 2.10: Weather protected transportation with temperature control and no humidity control

Class 2.10 is a combination of classes 2K11/2B2/2C2/2S5/2M4 in IEC 60721-3-2 [i.2].

This class shall apply to weather-protected transportation with limited temperature control and no humidity control and where special care has been taken (e.g. in respect to handling).

4.2 Class 2.11: Weather protected transportation with no temperature control and humidity control

Class 2.11 is a combination of classes 2K12/2B2/2C2/2S5/2M4 in IEC 60721-3-2 [i.2].

This class shall apply to weather-protected transportation without temperature and humidity controls in unventilated enclosures in arid, temperate, tropical and cold climates. Polar climate is excluded. The product may be transported in heated, pressurized aircraft holds.

4.3 Class 2.12: Non-weather protected transportation

Class 2.12 is a combination of classes 2K13/2B2/2C2/2S5/2M5 in IEC 60721-3-2 [i.2].

This class shall apply to non-weather-protected transportation in temperate, tropical, and cold climates. Arid and polar climates are excluded.

5 Environmental conditions

5.1 Climatic conditions

Table 1: Climate parameters for environmental classes 2.10, 2.11 and 2.12

			Classes		
	Environmental parameter	**Unit**	**2.10**	**2.11**	**2.12**
a)	Low temperature air	°C	-5	-45	-45
b)	High temperature, air	°C	+40	+70	+45 (see note 8)
c)	Low relative humidity (see note 1)	%	5	4	4
d)	High relative humidity (see note 1)	%	85	100	100
e)	Low absolute humidity (see note 1)	g/m^3	1	0,5	0,02
f)	High absolute humidity (see note 1)	g/m^3	25	29	35
g)	Rate of change temperature (see note 2)	°C/min	0,5	1	1
h)	Low air pressure (see note 3)	kPa	70	70	70
i)	High air pressure (see note 3)	kPa	106	106	106
j)	Solar radiation	W/m^2	No	See note 7	1 090
k)	Heat radiation	Not specified	None	None	None
l)	Movement of surrounding air (see note 4)	m/s	1,0	1,0	See note 6
m)	Condensation	Not specified	No	Yes	Yes
n)	Precipitation (rain, snow, hail, etc.)	Not specified	None	None	Yes
o)	Rain intensity	mm/min	None	None	3,3
p)	Driving rain	m/s	None	None	See note 6
q)	Snow load	Kg/m^2	None	None	100
r)	Low rain temperature (see note 5)	°C	None	None	+5
s)	Water from sources other than rain	Not specified	No	Dripping water	See note 9
t)	Formation of ice and frost (including freeze-thaw)	Not specified	No	Yes	Yes
u)	Temperature shock	Not specified	No	Yes	Yes

NOTE 1: The low and high relative humidity levels are limited by the low and high absolute humidity, so that, for example, for environmental parameters a) and c), or b) and d), the severities given in this table do not occur simultaneously.
NOTE 2: Averaged over a period of time of 5 min.
NOTE 3: The value of 70 kPa represents a limit for open-air conditions, normally at an altitude of 3 000 m. In some geographical areas, open-air conditions may occur at higher altitudes. Conditions in mines are not considered.
NOTE 4: A cooling system based on non-assisted convection may be disturbed by adverse movement of surrounding air.
NOTE 5: This rain temperature should be considered together with high air temperature b) and solar radiation j). The cooling effect of the rain has to be considered in connection with the surface temperature of the product.
NOTE 6: If applicable, a special value should be selected based on expected transportation mode (e.g. lorry, open decks of ships).
NOTE 7: Thermal effect of solar radiation is included in the temperature.
NOTE 8: Thermal effect of solar radiation is not included in the temperature.
NOTE 9: Sources of water other than rain are encompassed by driving rain.

5.2 Biological conditions

Table 2: Biological conditions for environmental classes 2.10, 2.11 and 2.12

	Environmental parameter	Unit	Classes 2.10 to 2.12
a)	Flora	none	Presence of mould, fungus, etc.
b)	Fauna	none	Presence of rodents or other animals harmful to products but excluding termites

5.3 Chemically active substances

Table 3: Chemically active substances for environmental classes 2.10, 2.11 and 2.12

	Environmental parameter	Unit (see note 1)	Classes 2.10, 2.11 and 2.12 (see note 2)
a)	Salt mist (including sea and road salt)	none	sea and road salt mist
b)	Saltwater	none	No
c)	Sulphur dioxide (SO_2)	mg/m^3 cm^3/m^3	1,0 0,37
d)	Hydrogen sulphide (H_2S)	mg/m^3 cm^3/m^3	0,5 0,36
e)	Hydrogen chloride (HCl)	mg/m^3 cm^3/m^3	0,5 0,33
f)	Hydrogen fluoride (HF)	mg/m^3 cm^3/m^3	0,03 0,036
g)	Ammonia (NH_3)	mg/m^3 cm^3/m^3	3,0 4,2
h)	Ozone (O_3)	mg/m^3 cm^3/m^3	0,1 0,05
i)	Nitrogen oxides (expressed as equivalent values of nitrogen dioxides) (NO_X)	mg/m^3 cm^3/m^3	1,0 0,52
NOTE 1: The values given in cm^3/m^3 have been calculated from the values given in mg/m^3 and refer to 20 °C and 101,3 kPa. The table uses rounded values. NOTE 2: The figures given are limit or peak values, occurring over a period of time of not more than 30 minutes per day.			

5.4 Mechanically active substances

Table 4: Mechanically active substances for environmental classes 2.10 2.11 and 2.12

	Environmental parameter	Unit	Classes 2.10, 2.11 and 2.12
a)	Sand in air	mg/m^3	30
b)	Dust in air	mg/m^3	0,2
c)	Dust sedimentation	$mg/(m^2h)$	15

5.5 Mechanical conditions

Table 5: Mechanical conditions for the environmental classes 2.10, 2.11 and 2.12

	Environmental parameter	Unit	Class								
			2.10			2.11			2.12		
a)	Stationary vibration, random										
	acceleration power spectral density	m^2/s^3	10 (see note 2)	1	0,5	10 (see note 2)	1	0,5	30 (see note 2)	3	1
	frequency range (see note 1)	Hz	2 to 10	10 to 20	20 to 2 000	2 to 10	10 to 20	20 to 2 000	2 to 10	10 to 20	20 to 2 000
b)	Non-stationary vibration, including shock (see note 3)										
	Shock 1 (see note 4)		Figure 2, curve 4 of IEC 60721-3-2 [i.2]			Figure 2, curve 4 of IEC 60721-3-2 [i.2]			Figure 2, curve 3 of IEC 60721-3-2 [i.2]		
	Peak acceleration	m/s^2	100			100			300		
	Duration	ms	11			11			11		
	Shock 2 (see note 4)		Figure 2, curve 2 of IEC 60721-3-2 [i.2]			Figure 2, curve 1 of IEC 60721-3-2 [i.2]			Figure 2, curve 3 of IEC 60721-3-2 [i.2]		
	Peak acceleration	m/s^2	300			1 000			300		
	Duration	ms	6			6			11		
c)	Free fall:		see note 8								
	mass < 20 kg	m	0,25			0,25			1,2		
	mass 20 kg to 100 kg	m	0,25			0,25			1,0		
	mass > 100 kg	m	0,1			0,1			0,25		
d)	Toppling										
	mass < 20 kg	none	Toppling around any of the edges			Toppling around any of the edges			Toppling around any of the edges		
	mass 20 kg to 100 kg	none	No			No			Toppling around any of the edges		
	mass > 100 kg	none	No			No			No		
e)	Rolling, pitching:										
	Angle (see note 5)	Degree	No			No			±35		
	Period	s	No			No			8		
f)	Steady state acceleration	m/s^2	20			20			20		
g)	Static load	kPa	5			5			10		

NOTE 1: When transport only occurs by rail, river, sea and road, the upper frequency considered may be reduced to 500 Hz for products that are not sensitive to vibration excitations above 500 Hz.
NOTE 2: The low frequency stationary vibration random component arises from the influence of land vehicle suspension systems. The component is included for design purposes but is not always included in vibration test specifications.
NOTE 3: For land vehicles, these shocks can occur simultaneously with the stationary vibration random conditions.
NOTE 4: Both shocks would normally be used to encompass different aspects of the shock environment.
NOTE 5: An angle of 35° may only occur temporarily. An angle of up to 22,5° can be reached for long periods of time.

Annex A (informative): Bibliography

- ETSI TR 100 035 (2004): "Equipment Engineering (EE); Environmental engineering; Guidance and terminology".

Annex B (informative): Change history

Version	Information about changes
3.1.1	The environmental classes have been revised based on the classification given in IEC 60721-3-2:2018. As the new classes are not comparable with the previous classes taken from IEC 60721-3-2:1997, the new environmental classes have been defined with new numbers.

History

Version	Date	Status	
Edition 1	February 1992	Publication as ETS 300 019-1-2	
V2.1.2	April 2002	Publication	
V2.1.4	April 2003	Publication	
V2.2.1	April 2014	Publication	
V3.1.0	June 2026	ENAP Process	AP 20260920: 2026-06-22 to 2026-09-21
V3.1.1	September 2026	Publication	